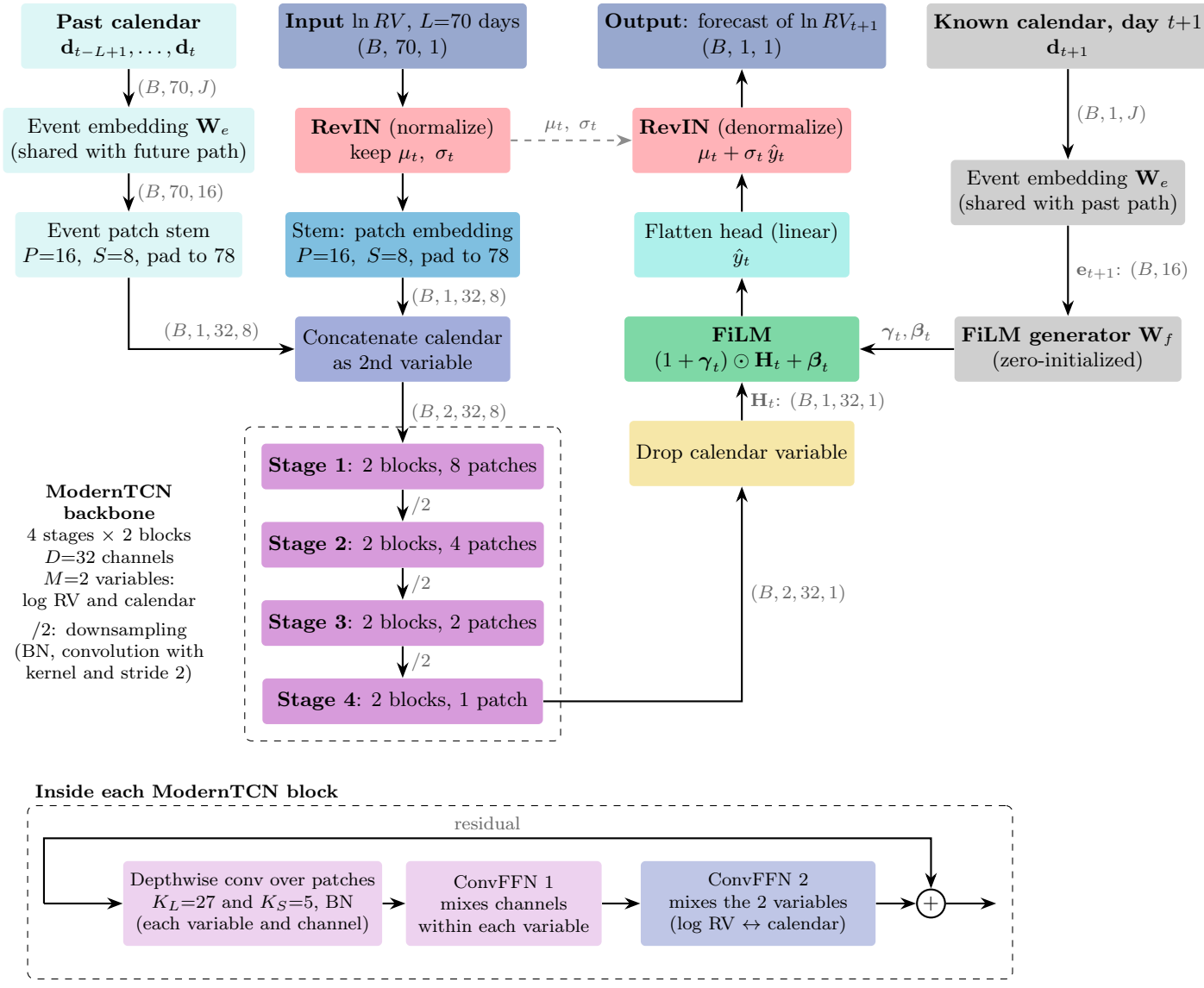

Past calendar
$\mathbf{d}_{t-L+1}, \ldots, \mathbf{d}_t$
$(B, 70, J)$
Event embedding $\mathbf{W}_e$
(shared with future path)
$(B, 70, 16)$
Event patch stem
P=16, S=8, pad to 78
$(B, 1, 32, 8)$
Input ln RV, L=70 days
$(B, 70, 1)$
RevIN (normalize)
keep μ_t, σ_t
μ_t, σ_t
Stem: patch embedding
P=16, S=8, pad to 78
$(B, 1, 32, 8)$
Concatenate calendar
as 2nd variable
$(B, 2, 32, 8)$
Stage 1: 2 blocks, 8 patches
/2
Stage 2: 2 blocks, 4 patches
/2
Stage 3: 2 blocks, 2 patches
/2
Stage 4: 2 blocks, 1 patch
ModernTCN
backbone
4 stages × 2 blocks
D=32 channels
M=2 variables:
log RV and calendar
/2: downsampling
(BN, convolution with
kernel and stride 2)
Output: forecast of ln RV_{t+1}
$(B, 1, 1)$
RevIN (denormalize)
$\mu_t + \sigma_t \hat{y}_t$
Flatten head (linear)
$\hat{y}_t$
FiLM
$(1 + \boldsymbol{\gamma}_t) \odot \mathbf{H}_t + \boldsymbol{\beta}_t$
$\boldsymbol{\gamma}_t, \boldsymbol{\beta}_t$
$\mathbf{H}_t$: $(B, 1, 32, 1)$
Drop calendar variable
$(B, 2, 32, 1)$
Known calendar, day t+1
$\mathbf{d}_{t+1}$
$(B, 1, J)$
Event embedding $\mathbf{W}_e$
(shared with past path)
$\mathbf{e}_{t+1}$: $(B, 16)$
FiLM generator $\mathbf{W}_f$
(zero-initialized)
Inside each ModernTCN block
residual
Depthwise conv over patches
K_L=27 and K_S=5, BN
(each variable and channel)
ConvFFN 1
mixes channels
within each variable
ConvFFN 2
mixes the 2 variables
(log RV ↔ calendar)